

Programming for Essential Digital Skills, Chapter 4

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Assigning Values to Variables

- We can assign *values* to *variables* using *assignment*.
- For example:

```
a = 2  
b = 3
```

- a and b are then visible in the **Variable Explorer** tab in Spyder.
- We can then use a and b for calculations:

```
a + b
```

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- There are rules for naming variables:
 - ▶ They must start with either a letter or `_` and contain no spaces.
 - ▶ Variable names cannot be a keyword in Python.

Data Types for Single Values

- Integers (`int`): whole numbers, positive or negative (e.g. 1 or -2).
- Floating-Point Numbers (`float`): real-valued numbers (e.g. 1.234).
 - ▶ Be careful with floating-point precision!
- Strings (`str`): sequences of characters (e.g. `'abc'`).
 - ▶ Addition combines strings: `'abc' + 'def'`
 - ▶ Multiplication by an integer repeats strings: `3 * 'abc'`
- Boolean (`bool`): The Logical constants `True` and `False`.

Logical Operators

There are 3 logical operators: and, or and not:

- a and b is True if both a and b are True. Otherwise it is False.
- a or b is True if either a or b are True. Otherwise it is False.
- not a is True if a is False. Otherwise it is True.

```
a = True  
b = False  
a and b
```

False

```
a or b
```

True

```
not a
```

False

Comparison Operators

We can also check weak/strict (in)equalities in Python:

```
a = 2  
b = 3  
a == b  # is a equal to b?
```

False

```
a != b  # is a not equal to b?
```

True

```
a > b  # is a strictly greater than b?
```

False

```
a <= b  # is a weakly less than b?
```

True

Type Conversion

We can sometimes convert between `int`, `float`, `str` and `bool`, but we don't always get intuitive results!

```
float('1.0')
```

1.0

```
str(1)
```

'1'

```
int(True)
```

1

```
bool('False')
```

True